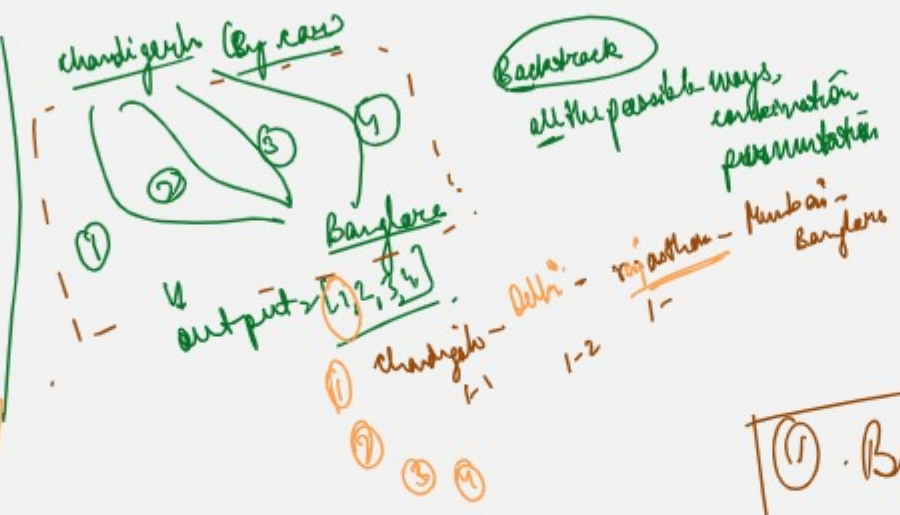


Backtrack
 $(n)(n-1)(n-2) \dots 1$

Permutation, target value
 $= (N \cdot N!)$

$(n! \cdot \log n)$
 $(2^n \cdot \log n)$

Time complexity Backtrack
 Subsequent's Problem Hamiltonian
 $(2 \cdot N)$
 (combination)
 all state



Greedy algo (specify)

Strategy

Weights
 Time interval, Best combination for profit, max. nize. profit, value.

Sort a array

Activity taken

Knapsack
 something, room, schedule, event, organizing, intervals

Car, b
 Car, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z

elements
 (constant, bag and fit in elements max. value - weight)
 (price)

Identify the parameters (min, weight, price)

Sort phase

Looping and initializing

Start over sorted array and check for the value.

if else, current item is making the condition, look it in

Buy and sell stock (multiple buy and sell) (maximum profit)
 stock
 also $= [7, 1, 5, 3, 6, 2]$
 price $= 4 + 3 = 7$

greedy

Meeting Intervals

start $= [10, 8, 9, 11.5]$ (min represent)
 end $= [11, 9.5, 12, 13]$ (min represent)
 conference room
 [maximum no. of meeting possible today]
 not overlapping

$[10, 11]$
 $[8, 9.5]$
 $[11.5, 12]$

Bag - 50kg
 Pick value
 4 saffron - 50kg (\$600)
 4 vanilla - 20kg (\$1000)
 4 cinnamon - 30kg (\$1200)
 $\frac{1200}{20} = 60$
 $\frac{1000}{20} = 50$
 $\frac{600}{20} = 30$
 60kg.
 \$1000 - 20kg
 \$600 - 10kg
 \$800
 2200\$ (maximum)

